

Optical Switch & Counter

Laboratory Experiment #8

By

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Submitted To:

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EST 2221C – Electro-Optical Devices
LABORATORY #7: Optical Switch & Counter

Est. Time Required: 1:00 – 1:30 hrs.

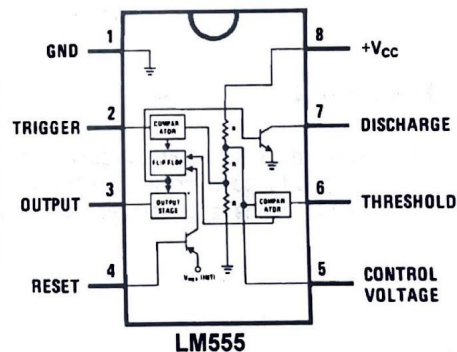
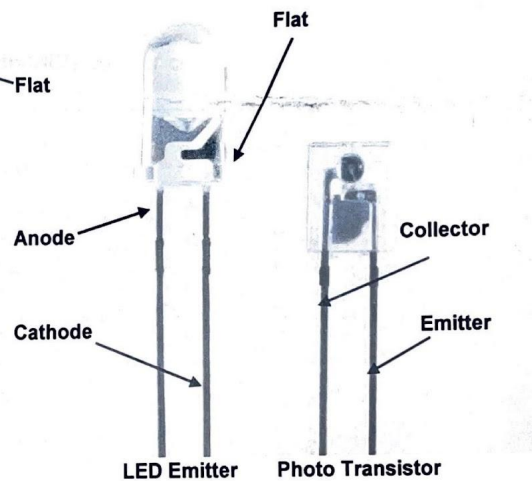
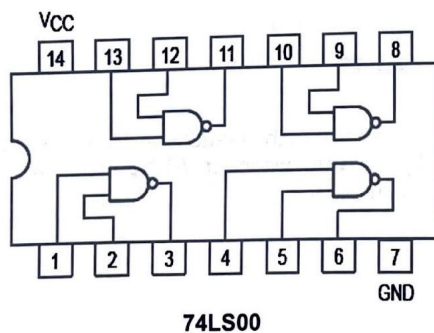
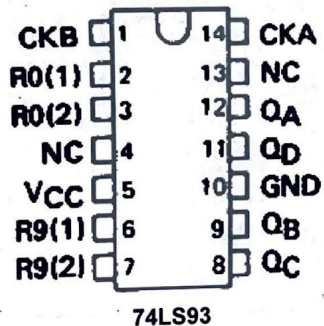
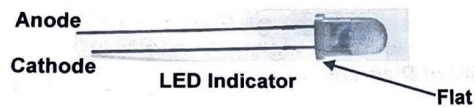
OBJECTIVE:

- To incorporate optical components into electronic circuits.

MATERIALS:

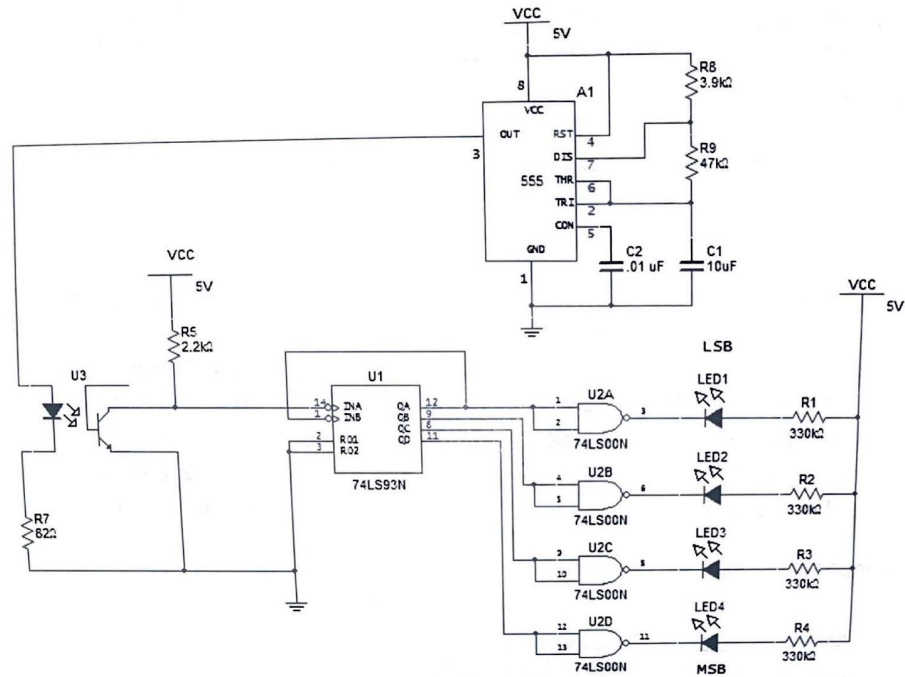
- 5 Vdc power supply
- 74LS93N four bit counter
- 555 Timer
- 74LS00 Quad NAND Gate
- 4x Red /Green /Yellow Indicator LEDs
- Emitter LED
- Photo Diode
- 0.01uF Capacitor
- 10uF Capacitor
- 82 Ω Resistor
- 4x 330 Ω Resistor
- 2.2 K Ω Resistor
- 3.9 K Ω Resistor
- 47 K Ω Resistor

Component Notes:

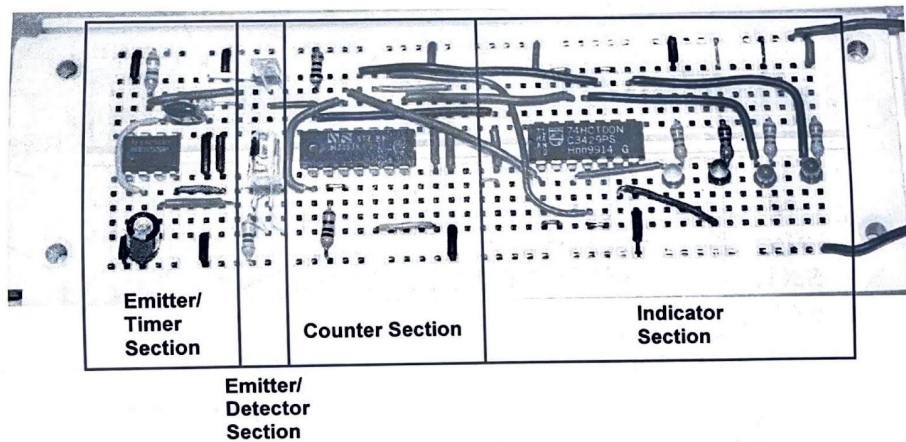


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Circuit Diagram



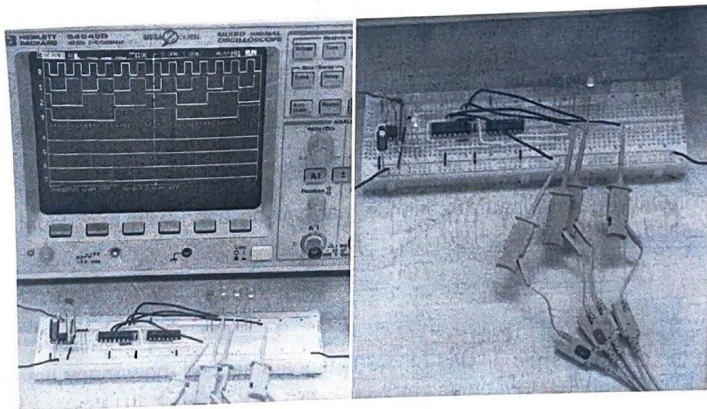
PROCEDURE:

1. Assemble the emitter section first on the left side of the breadboard from the above diagram. It is a good idea to color code the wires by using red wires for VCC connections, black for ground connections, and white/yellow for interconnections.

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2. Wire the LED so that the output faces the location where the detector (Phototransistor) will be located which includes bending the leads.
3. Connect a 5Vdc power supply to the VCC and Ground.
4. Power ON the power supply.
5. Verify that the 555 emitter circuit is functioning by observing the emitter LED. The emitter LED should blink on and off about once a second (on for half a second then off for half a second).
6. Place the photo detector about 1cm away from the emitter.
7. Align the photo detector so that the light emitted by the emitter enters the detector.
8. Power OFF the power supply.
9. Assemble the counter and indicator circuits on the right side of the bread board. Note that the power supply pins on the 74LS93 are not in the usual location (corners).
10. Power ON the power supply.
11. Verify that the indicator LEDs light up in a binary sequence for each time the emitter turns off.
12. If the indicator LEDs turn on and off, however, the count is random or out of order then remove R5 (2.2K Ω) from the circuit.
13. If the emitter lights but the counter does not count then place a temporary jumper between pin 3 of the 555 timer and pin 14 of the 74LS93. If the counter still does not work, troubleshoot the counter circuit and remove the jumper. If the counter counts only with the jumper in place then remove the jumper and reposition the emitter and detector or troubleshoot the emitter and detector.



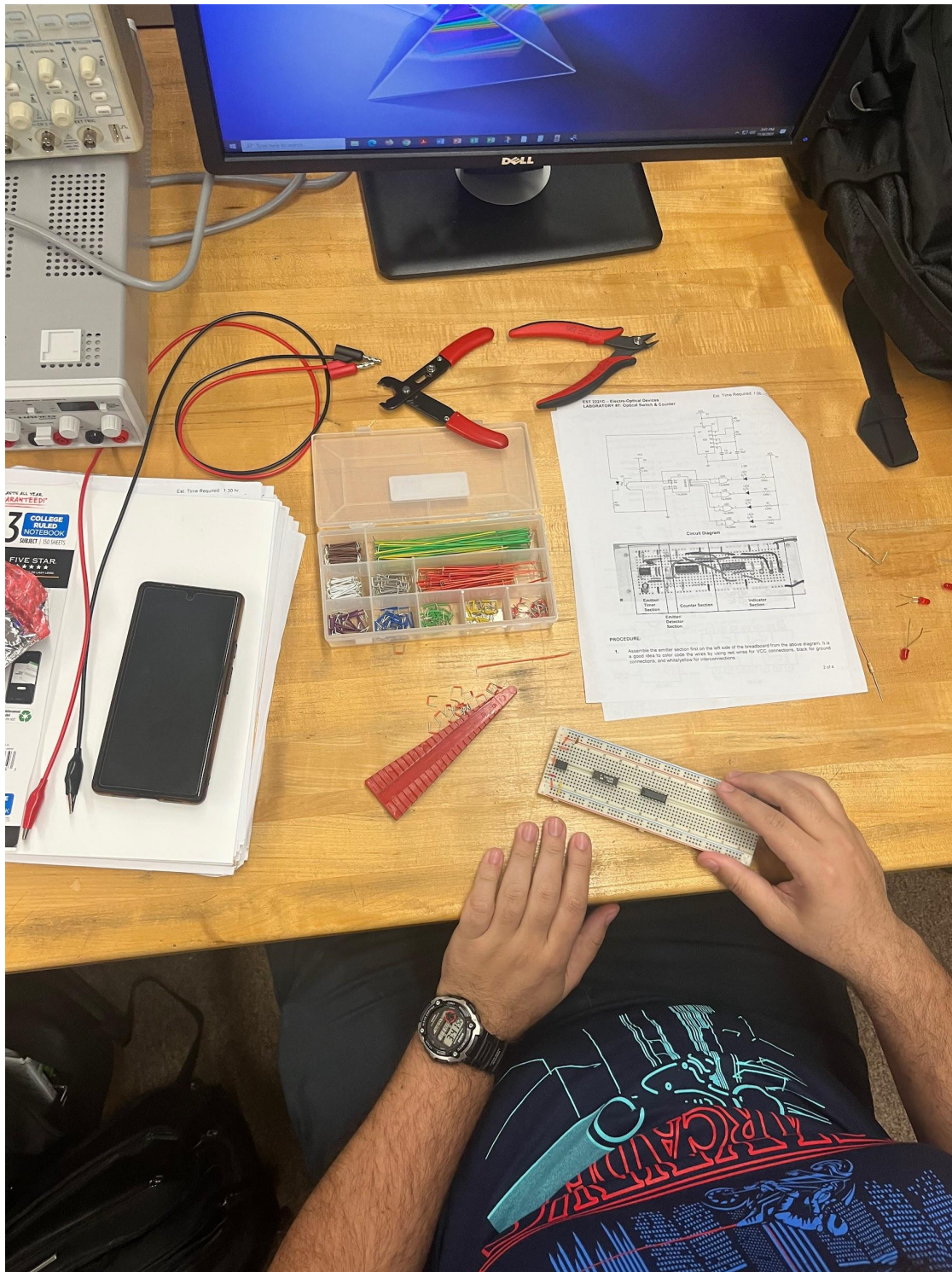
Circuit connected to the Oscilloscope

14. Insert the logic analyzer probe to the oscilloscope. Insert test wires to pins 8,9,11, and 12 of the 74LS93.
15. Connect channel 0 to pin 12 of the 74LS93.
16. Connect channel 1 to pin 9 of the 74LS93.
17. Connect channel 2 to pin 8 of the 74LS93.
18. Connect channel 3 to pin 11 of the 74LS93.
19. Power ON the oscilloscope.

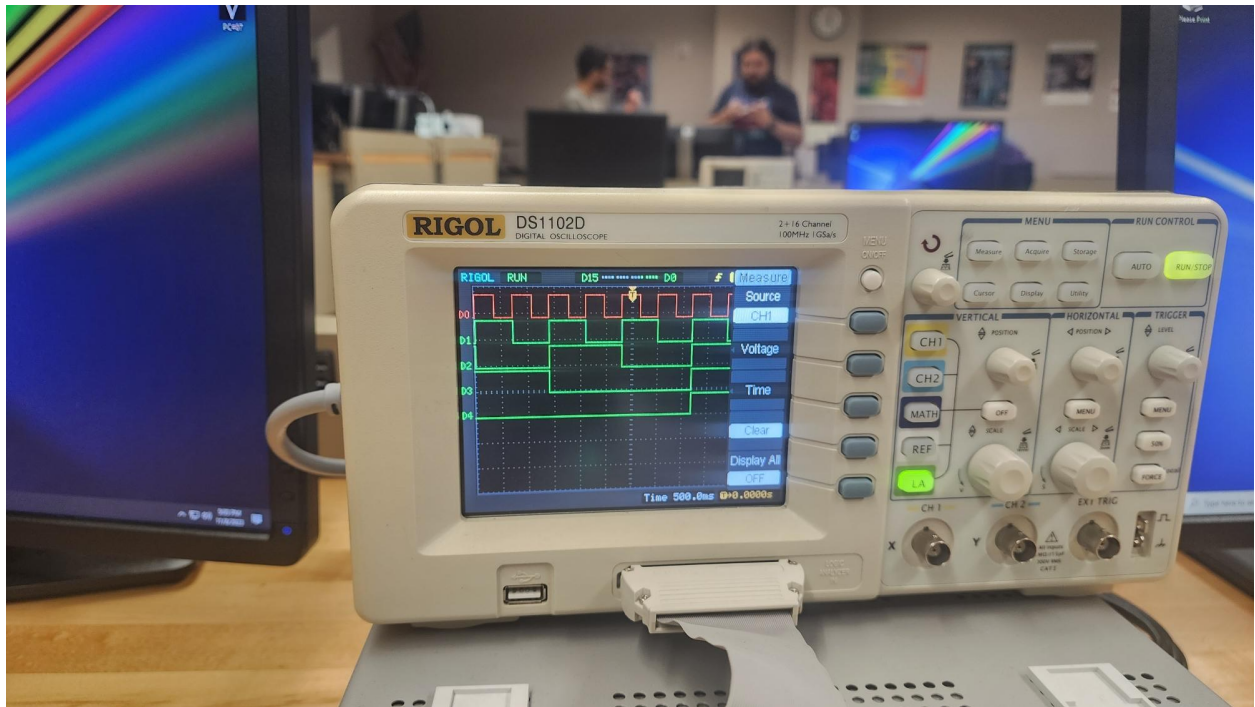
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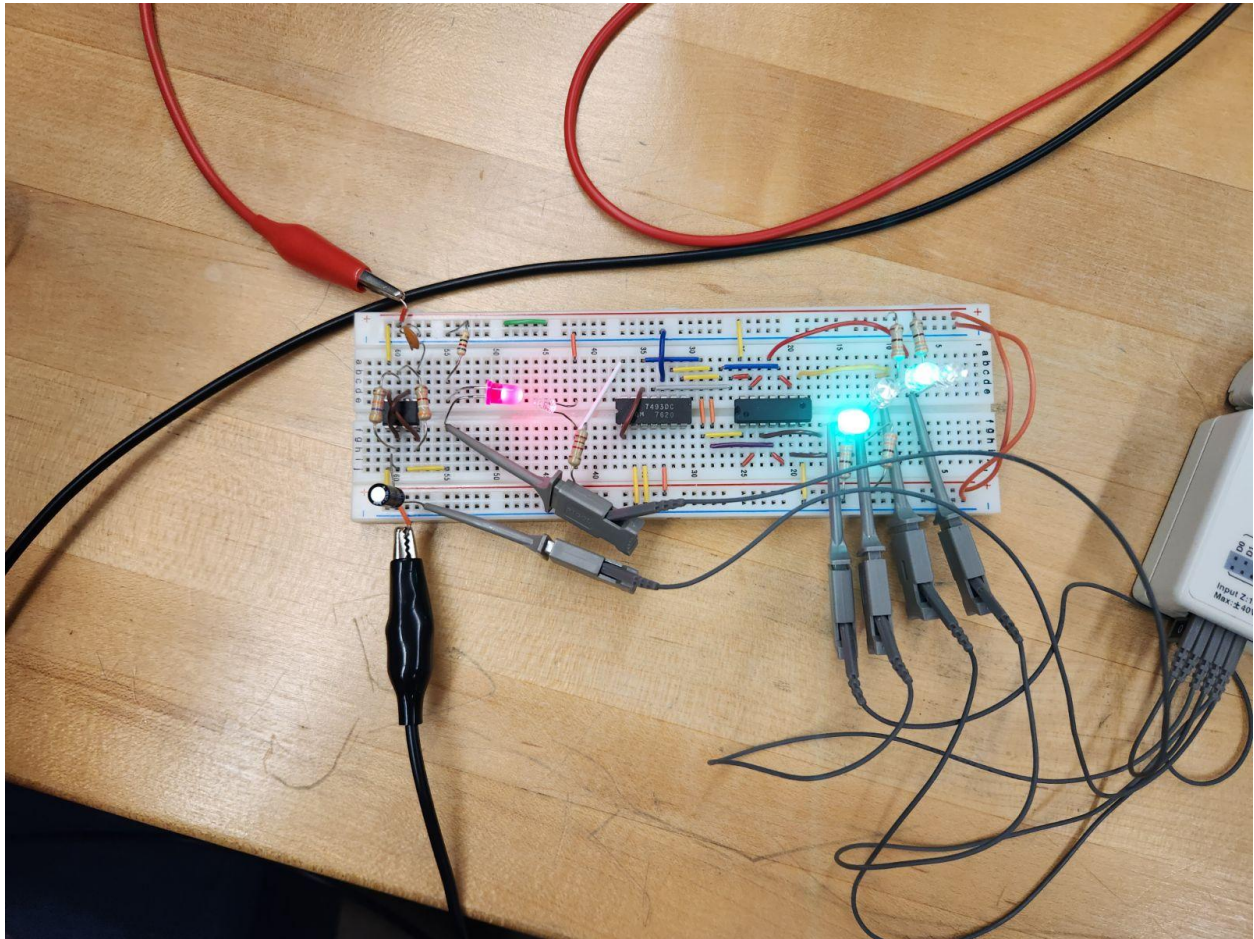
Est. Time Required: 1:00 – 1:30 hrs.

20. Press the D0-D15 button of the oscilloscope to set the display to view D0-D7 only.
21. What type of counter is indicated by the LEDs? up
22. What is the MOD number? 16
23. Complete the following table:



Lab setup. Making sure the LEDs and chips are set up properly.





The LEDs shown on the oscilloscope in different square waves according to the LED.

5.) Data & Observations

TABLE 1 Observations

Count	Q0	Q1	Q2	Q3	HEX
0	0	0	0	0	0
1	0	0	0	1	1
2	0	0	1	0	2
3	0	0	1	1	3

4	0	1	0	0	4
5	0	1	0	1	5
6	0	1	1	0	6
7	0	1	1	1	7
8	1	0	0	0	8
9	1	0	0	1	9
10	1	0	1	0	10
11	1	0	1	1	11
12	1	1	0	0	12
13	1	1	0	1	13
14	1	1	1	0	14
15	1	1	1	1	15

Calculations

The data obtained is as accurate as possible by measurement tools and the conductors of the experiment. This data shows the different colored LEDs have flip flops at the maximum frequency. Some errors can occur during the experiment if the laser, chopper, and power detector aren't aligned or the measured distance and calculations are incorrect.

6.) Answers to Lab Questions

24. Calculate the maximum frequency that can be used with this counter ($t_{pd} = 46 \text{ ns}$ & $N =$ the number of FFs).

$$f_{\max} = \frac{1}{N \cdot t_{pd}} = \frac{1}{46.4} \quad f_{\max} = 0.00543 \text{ Hz}$$

25. If the emitter turned on and off 1,600 times a second, how many times a second would the last LED (Channel 3) turn on and off?

1,600 $\frac{1}{s}$ $\rightarrow$ 1,600 Hz } 800 times per sec
 DC = 50%

7.) Conclusion

In conclusion, the laboratory experiment showed that the flip flops were affected by the maximum frequency. The experiment yielded expected results because the oscilloscope displayed the appropriate results respected to the individual wavelengths. Improvements would be to ensure that user errors and equipment errors are at a minimum so that the results are as accurate as possible for future work.